$$\frac{1}{2}mu_0^2 - G\frac{M_e m}{R_e} = \frac{1}{2}mu^2 \ . \qquad (13)$$

The surface gravitational acceleration is

$$g = G\frac{M_e}{R_e^2} \ , \qquad (14)$$

Solving (13), (14) gives

$$u_0 = \sqrt{u^2 + 2gR_e} \ . \qquad (15)$$

From (15) and the data, we obtain

$$u_0 = 40.7 \text{ km/s} \ . \qquad (16)$$

If the craft enters at E in the opposite direction, (11) must be changed. It then reads

$$u = \sqrt{v^2 + v_e^2 - 2vv_e\cos(\frac{\pi}{2} + \theta)} \ , \qquad (17)$$

This gives the other solution:

$$u = 45.0 \text{ km/s} \ , \quad u_0 = 46.4 \text{ km/s} \ . \qquad (18)$$

For entry at the point symmetric to E (the other intersection of Earth's and the craft's solar orbits), the calculation and results are unchanged.

3.

The two springs in series have an effective spring constant

$$k = \frac{k_1 k_2}{k_1 + k_2} \ . \qquad (1)$$

Below A is $v$mol gas. For A at height h_1: pressure p_1, temperature T_1. ideal-gas equation and equilibrium condition give

$$p_1 S h_1 = vRT_1 \ , \qquad (2)$$

$$p_1 S = kh_1 + mg \ . \qquad (3)$$

After heating, for A at height h_2: pressure p_2, temperature T_2. The ideal-gas law and equilibrium condition give

$$p_2 S h_2 = vRT_2 \ , \qquad (4)$$

$$p_2 S = kh_2 + mg \ . \qquad (5)$$

As A rises from height h_1 to h_2, the increase in the gas's internal energy is

$$\triangle U = v\frac{3}{2}R(T_2 - T_1) \ . \qquad (6)$$

Gas work W equals gains in spring elastic and piston gravitational energy: